Data Analytics Project: Railway Gauge Trends

Dataset Used: **Railwaygauge_1**

Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.Modules Used.

The primary libraries/modules required for this project are:

● Pandas: For all data manipulation, filtering, aggregation, and feature engineering tasks.

● NumPy: Specifically for numerical calculations like yearly growth (np.diff).

● Visualization Library (e.g., Matplotlib): For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

Category	Skills Covered
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)

Visualization	**Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart**

Project 1 **Title: Railway_Gauages**

Scenario 1:

Basic Data Loading & Cleaning:

The mainly steps of working with a dataset using Pandas and data loading is the process of importing data from a file for analysis. When data is incomplete by cleaning the data .we make it for further analysis visualization.

1.Load the dataset into a Pandas DataFrame: Pandas provides functions like read_csv() to data load into a DataFrame and it easy to analyze data.

```
#1. Load the dataset into a Pandas DataFrame.
df=pd.read_csv("Railwaygauge_1.csv")
```

2. **Display the first 5 rows and column names to understand structure:** what your dataset looks like loading it into Pandas. It helps you understand the structure, columns, and values before doing any analysis.

```
#2.display the first 5 rows and column names.
print("top 5 Rows are\n",df.head(),'\n')
```

3. **Check for missing values and replace them with 0** : Missing values are replaced with 0 to avoid error .

```
#3.. Check for missing values and replace them with 0.
# mising values check for column
print("missing values are\n",df.isnull().sum(),'\n')
#returns
missing_row=df[df.isnull().any(axis=1)]
print(missing_row)
#replaceing missing values with 0

print(df)
```

4. **Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types:**

Columns like Broad, Metre, Narrow, Total are converted to numbers so we can do run.

```
#4. Convert all gauge columns (Broad, Metre, Narrow, Total) to numeric types.
df["Broad Gauge"]=pd.to_numeric(df["Broad Gauge"])
df["Metre Gauge"]=pd.to_numeric(df["Metre Gauge"])
df["Narrow Gauge"]=pd.to_numeric(df["Narrow Gauge"])
df["Total"]=pd.to_numeric(df["Total"])
print(df.info())
```

Conclusion:

Data cleaning is very important before analysis. It removes errors and makes data easy to run. After cleaning, the dataset becomes easy to use. This helps in getting correct results. Clean data gives better output.

Scenario 2:

Simple Visualization

we create a graph to understand data easily. We use Year and Total columns. A line graph helps us see how railway tracks change over time. Graphs make data simple and easy to understand**.**

1. **Extract the Year and Total columns:** Select the data take only year and Total columns.

```
#2.1 Extract Year and Total columns.
df=pd.read_csv("railwaygauge_1.csv")
print(df[['Total','Year']])
```

2. **Plot a line graph showing Total tracks over the Years**: A line graph show the everytime changes.

```
#2.2 Plot a line graph showing Total tracks over years.
plt.plot(df['Year'],df['Total'])
plt.xticks(rotation=70)
```

3. **Add:**
   - **Title**
   - **X and Y labels**

Title explain the graph

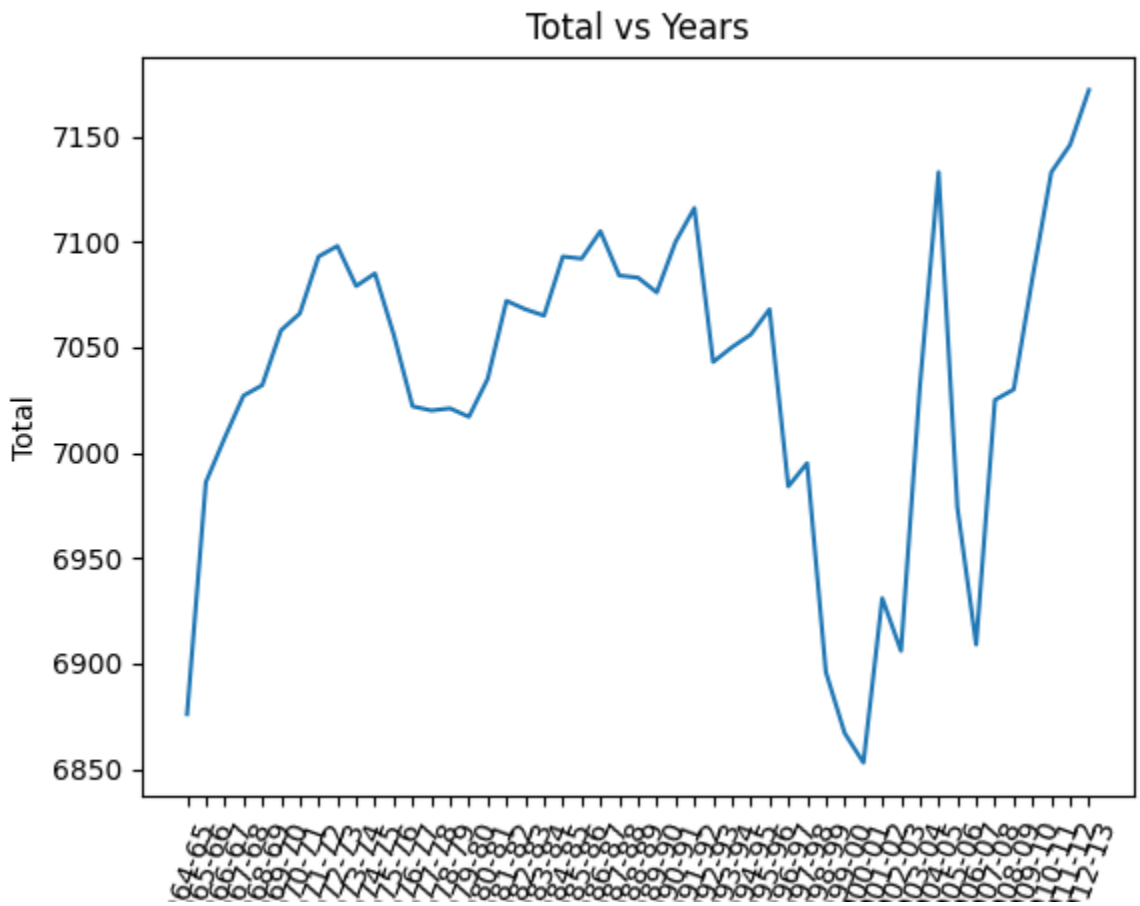
- x-axis=year
- y-axis=Total

```
#2.3 Add: Title X and Y labels
plt.xlabel('Year')
plt.ylabel('Total')
plt.title('Total vs Years')
plt.savefig("total_over_year.png")
```

4. **Identify whether the overall trend is increasing or decreasing**: A line is too straight

   When track is up its increasing and track is down its decreasing.

```
#2.4 Identify whether the trend is increasing or decreasing.
plt.show()
"""
```



Conclusion: data makes easy to understand. A line graph clearly shows changes over time. It helps to find trends. This makes analysis simple.

Scenario 3:

Filtering + Bar chart

We knows to railway data We filter the dataset to after 2000 keep only recent years. Then we compare different railway gauges using a bar chart. A bar chart helps to multiple categories types. its easy to see which gauge is more used in recent years.

1. **Filter the dataset to include only years after 2000:** Select the only years after 2000 to run the data.

```
# Convert Year
df[year_col] = pd.to_numeric(df[year_col].astype(str).str[:4], errors='coerce')

# 1. Filter after 2000
df = df[df[year_col] > 2000]

# Check empty
if df.empty:
    print("No data after 2000")
else:
```

2. **Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns :**

Only used to selective column:

- Broad Gauge
- Metre Gauge

- Narrow Gauge

```
# 2. Select columns
df = df[[year_col, "Broad Gauge", "Metre Gauge", "Narrow Gauge"]]
```

3.Plot a grouped bar chart comparing the three gauges : Compare the all three gauges each year has three bar.

```
# 3. Plot grouped bar chart
df.set_index(year_col).plot(kind='bar')
```

4. Add a legend and ensure proper labels :

We add to :

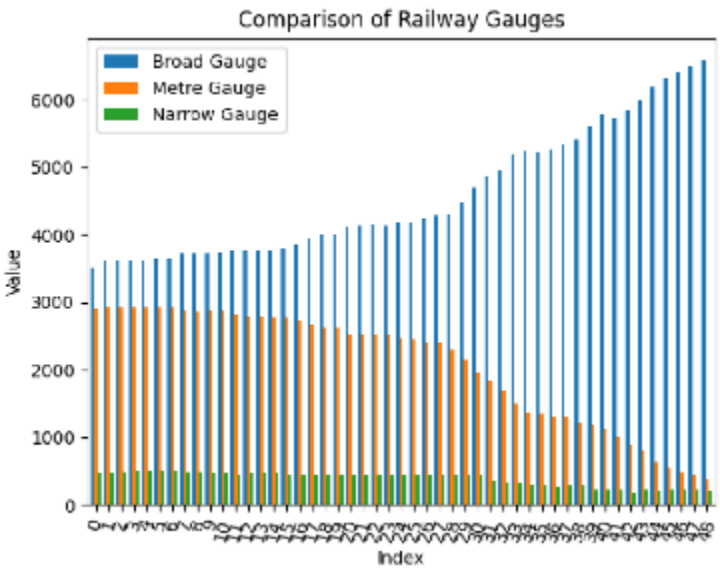
- Xlabel
- Ylabel
- Legend

```
# 4. Labels
plt.title("Comparison of Railway Gauges After 2000")
plt.xlabel("Year")
plt.ylabel("Track Length")
plt.legend()

# Save graph
os.makedirs("Graphs", exist_ok=True)
plt.savefig("Graphs/Broad_Metre_Narrow_bar.png")
```

5. **Identify which gauge dominates in recent years:** Dominating means the railway gauge that has the high value to other value.

```
# 5. Dominating gauge
total = df[["Broad Gauge", "Metre Gauge", "Narrow Gauge"]].sum()
print("Dominating gauge:", total.idxmax())
```



Conclusion:

Mainly focus on recent data. The bar chart makes comparison with graph we can show the which gauge is most used. This helps to railway development trends.

Scenario 4:

Feature Engineering + Pie Chart

1. . **Calculate total sum of each gauge across all years:** Add the values of each gauge(Broad,Metre, Narrow) all years.find the length of each gauge.

2. **Create a new structure (Series/DataFrame) for totals:** makes data easy to use for plotand organizes all total in one place.

3. **Plot a pie chart showing percentage contribution :**

   - Each slice = one gauge
   - Size=its contribution

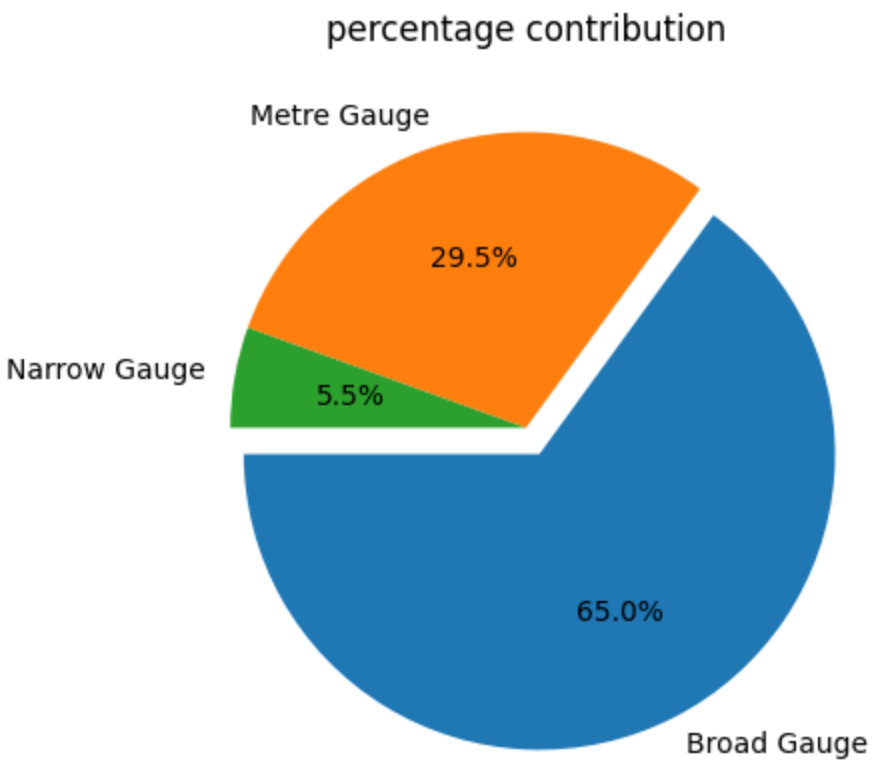
4. **Add percentage labels (autopct**):

   Autopct='%1.1f%%

   - Show the percentage on graph
   - Graph makes more clear

5.Interpret which gauge contributes the most:

- Largest slice=highest contribution
- Small slices=less contribution



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Conclusion: From the pie chart we can easily say that Broad Gauge have the most contribution for the tracks .

Scenario 5:

Advance analysis + multiple graphs:

1. Create new columns:

- **% Broad Gauge =** (Broad/total)*100
- **% Metre Gauge =** (Metre/total)*100
- **% Narrow Gauge=**(Narrow/total)*100

2. Use NumPy (np.diff) to calculate yearly growth of Total tracks:
np.diff() calculation the difference between years.
. measures year to year growth

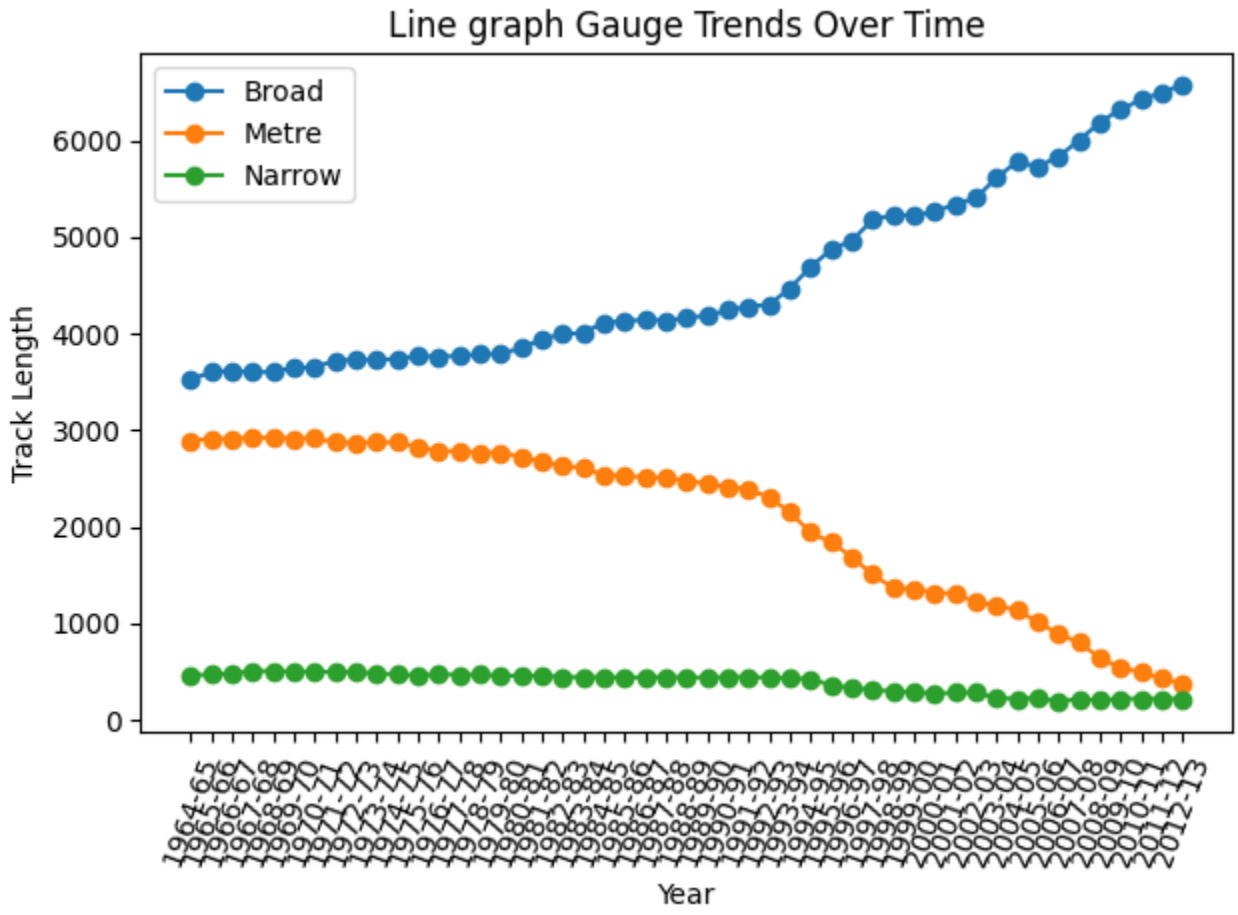
3. Plot:
○ **Line graph for all gauges :** Displays trends all time for all gauge..
○ **Stacked bar chart showing composition**: Shows total railway track length into gauge components.

4.Highlight:
○ **Years with highest growth :** mainly step on railway data is analyze the all time. increase the railway length.
○ **Decline in any gauge:** if any gauge is decreasing over time.

5.Provide a final conclusion:

"Is the railway system shifting towards a single dominant gauge?"

- Broad gauge→ increase
- Metre & Narrow→ decreasing



Conclusion:

Yes, the railway system is shifting towards a single dominant gauge (Broad Gauge).

Broad Gauge is increasing, while Metre and Narrow Gauges are decreasing.